Arista Access Node ELEC-ACC-01 Routing Protocol Flap SOP for Electronic City Hub

Node: ELEC-ACC-01 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_5633

Description

A routing protocol flap on Arista Access Node ELEC-ACC-01 at Electronic City Hub is a condition where the BGP route churn, OSPF adjacency loss, or ISIS LSP flooding triggers the ALARM_5633 (Severity: Warning) alarm code. This situation indicates a potential instability in the routing protocol operations, which may be caused by various factors such as misconfigured routing policies, suboptimal network design, or hardware-related issues. In an Arista Access Node, the routing protocol flap can occur due to the interactions between the control and data planes. When BGP, OSPF, or ISIS protocols operate at the control plane, they exchange routing information to establish and maintain routing tables. However, when these protocols experience issues, such as route flapping or adjacency loss, it can lead to an unstable routing environment. The data plane, which is responsible for forwarding packets, may also be affected by the routing protocol flap, resulting in packet loss or delay. To troubleshoot and resolve this issue, it is essential to identify the root cause of the routing protocol flap, which may involve analyzing the configuration, checking the hardware status, and verifying the routing protocol settings.

Pre-requisite Checks

- Item: Dispatch remote hands to Electronic City Hub to perform physical checks and verify the redundant link status.

- Item: Verify the Arista Access Node ELEC-ACC-01 is powered on and the console port is accessible.

- Item: Check the network topology and configuration to ensure there are no misconfigured routing policies or suboptimal network design.

Potential Root Causes

Cause	Probability
Misconfigured routing policies	High
Suboptimal network design	Medium
Hardware-related issues	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP summary information, Description: This command displays a summary of BGP sessions, including the number of established connections, total messages sent and received, and the number of routes received and accepted. It helps to identify if there are any BGP-related issues.
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Command: show ospf neighbor, Expected Output: OSPF neighbor information, Description: This command displays information about the OSPF neighbors, including their IP addresses, states, and hello intervals. It helps to identify if there are any OSPF adjacency issues.

Command: show isis database, Expected Output: ISIS database information, Description: This command displays the ISIS database, including the LSPs and their contents. It helps to identify if there are any ISIS-related issues.
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Resolution Steps

Step 1: Step: Check and correct the routing policies, Command: show running-config | include bgp, Description: This command displays the running configuration and checks if there are any misconfigured routing policies. If found, correct the policies to resolve the issue.

Step 2: Step: Verify the network design, Command: show ip route, Description: This command displays the routing table and checks if there are any suboptimal network design issues. If found, modify the network design to resolve the issue.

Step 3: Step: Perform a hardware reset, Command: reload, Description: This command reloads the hardware, which may resolve any hardware-related issues.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP summary information without any issues, Description: This command verifies if the BGP-related issues have been resolved.

Command: show ospf neighbor, Expected Output: OSPF neighbor information without any issues, Description: This command verifies if the OSPF adjacency issues have been resolved.

Command: show isis database, Expected Output: ISIS database information without any issues, Description: This command verifies if the ISIS-related issues have been resolved.
```

Roll-back Procedures

- Step: Restore the previous configuration, Command: copy running-config startup-config, Description: This command restores the previous configuration, which may resolve the issue.

- Step: Restore the previous network design, Command: copy running-config startup-config, Description: This command restores the previous network design, which may resolve the issue.

- Step: Reload the hardware, Command: reload, Description: This command reloads the hardware, which may resolve any hardware-related issues.